



Information Technology Institute (ITI)

Analog IC Design Track

Summer Camp 2026

Lab 5

Simple vs Low Compliance Cascode Current Mirror

Prepared by

Hussein Magdy Abdelalim Ahmed



Table of Contents

List of figures	4
Part 1: Sizing Chart	5
Nmos sizing.....	5
→ I_{dx} @ $V_{gs} = 605.378$ mV	6
→ Plot g_m , and g_{ds} vs V_G	6
Pmos sizing	7
→ I_{dx} @ $V_{gs} = 603.881$ mV	7
→ Plot g_m , and g_{ds} vs V_G	8
Design Specifications Summary	8
PART 2: Current Mirror	9
1) Schematic of the two CMs with DC node voltages clearly annotated at $V_{OUT} = V_{DD}/2$	9
2) Snapshots showing the following parameters for all transistors at $V_{OUT} = V_{DD}/2$	10
3) Check that all transistors have V_{GS} , g_m and g_{ds} as designed in Part 1.	11
4) Are all transistors operating in saturation?	11
5) I_{BOUT} vs V_{OUT} (DC Sweep).....	11
• Comment on the difference between the two circuits.....	12
• I_{BOUT} of the simple CM is exactly equal to $I_{BIN} \cdot 2$ at a specific value of V_{OUT} . Why?	12
6) Percent of Error in I_{BOUT} vs V_{OUT}	12
• Comment on the difference between the two circuits.....	13
7) R_{out} vs V_{OUT} (Log Scale)	13
• Comment on the difference between the two circuits.....	13
• Does R_{out} change with V_{OUT} ? Why?	13
8) Analytical vs Simulation Table ($V_{OUT} = V_{DD}/2$)	14



- Simple Current Mirror R_{out}14
- Wide-Swing R_{out} 14
- Table 2: Comparison of Analytical and Simulated R_{out} at $V_{OUT} = V_{DD}/2$ 14



List of figures

Figure 1: sizing schemtic	5
Figure 2: V^* & V_{ov} for nmos	5
Figure 3: I_{dx} vs. v_{gs} for nmos	6
Figure 4: g_m & g_{ds} vs. V_{gs}	6
Figure 5: V^* & V_{ov} for pmos	7
Figure 6: i_{dx} vs v_{gs} for pmos	7
Figure 7: g_m & g_{ds} for pmos	8
Figure 8: wide-swing current mirror	9
Figure 9: simple mirror	9
Figure 10: tb_schemtic	10
Figure 11: dc_op_wide_swing	10
Figure 12: dc_op_simple_mirror	11
Figure 13: $I_{BOU T}$ vs $V_{OU T}$ for wide swing cm	11
Figure 14: $I_{BOU T}$ vs $V_{OU T}$ for simple cm	12
Figure 15: errors calculation	12
Figure 16: R_{out_wcm} vs. R_{out_smplcm}	13



Part 1: Sizing Chart

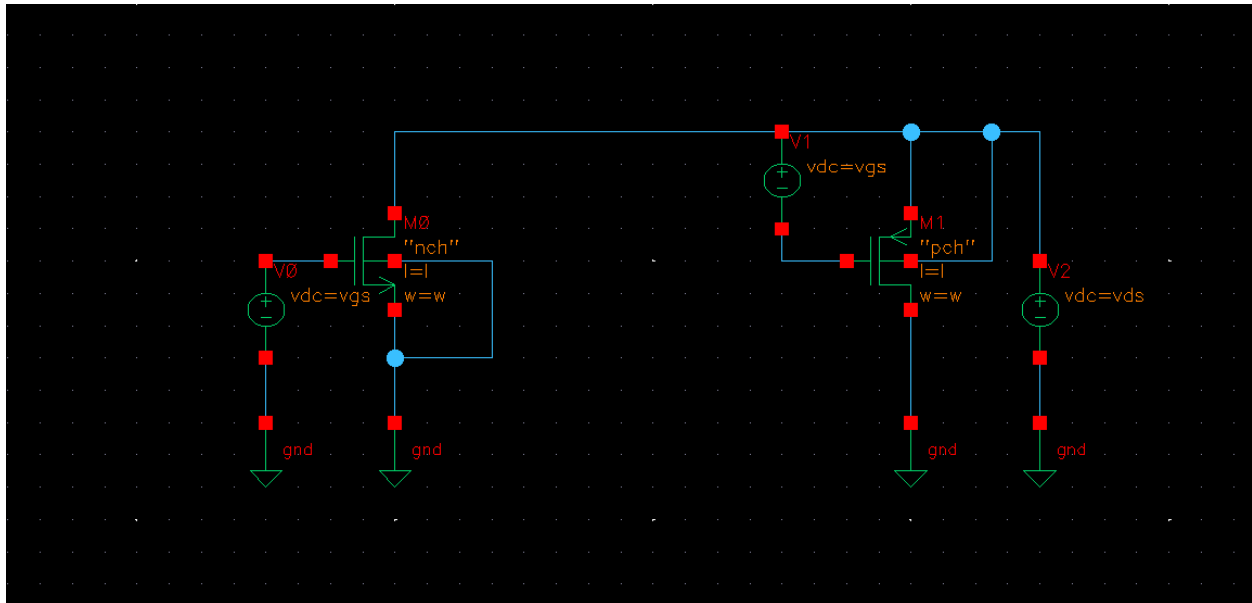


Figure 1: sizing schematic

→ We use $w_n = w_p = 10\mu\text{m}$, $L_n = L_p = 1\mu\text{m}$

Nmos sizing

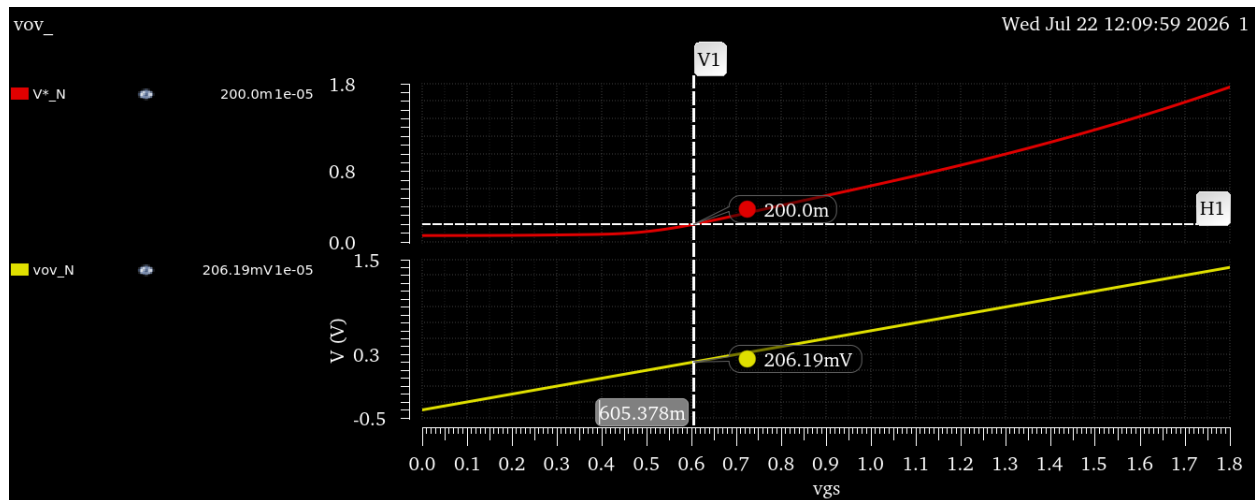


Figure 2: V^* & V_{ov} for nmos

- At $V^*=200\text{ mV} \rightarrow V_{gs}=605.378\text{ mV}$



→ I_{dx} @ $v_{gs} = 605.378 \text{ mV}$

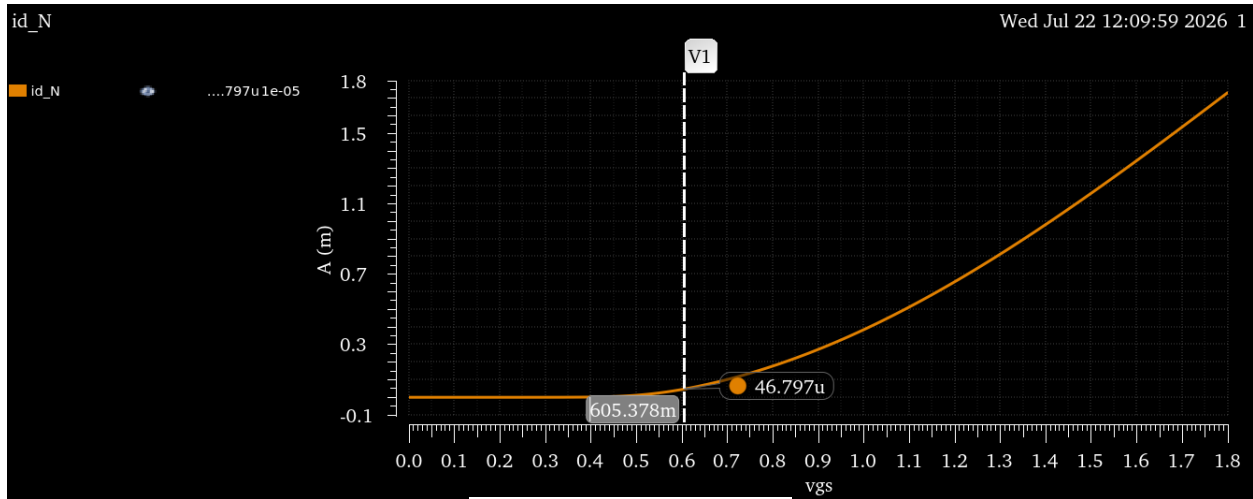


Figure 3: I_{dx} vs. v_{gs} for nmos

→ Plot g_m , and g_{ds} vs V_G

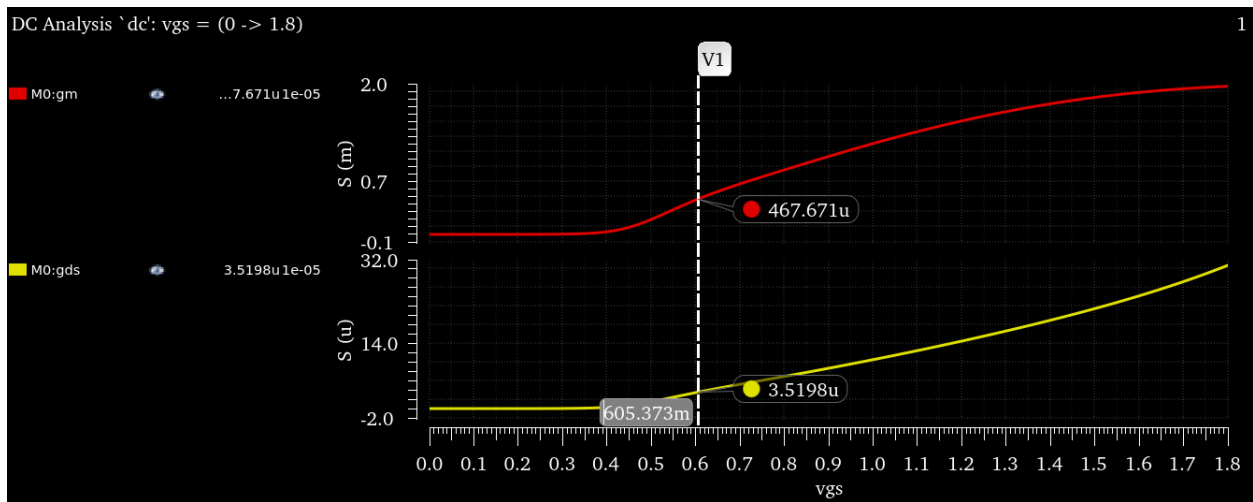


Figure 4: g_m & g_{ds} vs. V_{gs}

- At $V_{gs} = 605.378 \text{ mV}$, from graph $g_m = 467.671 \text{ uS}$ & $r_o = \frac{1}{g_{ds}} = 284.1 \text{ kohm}$
- $I_{dx} = 46.797 \text{ uA} \rightarrow W = 10 \text{ um}$
- $I_d = 20 \text{ uA} \rightarrow W_n = ?$
- $W_n = 10 \text{ um} \times \frac{20 \text{ uA}}{I_{DX}} = 4.2737 \text{ um}$



Pmos sizing

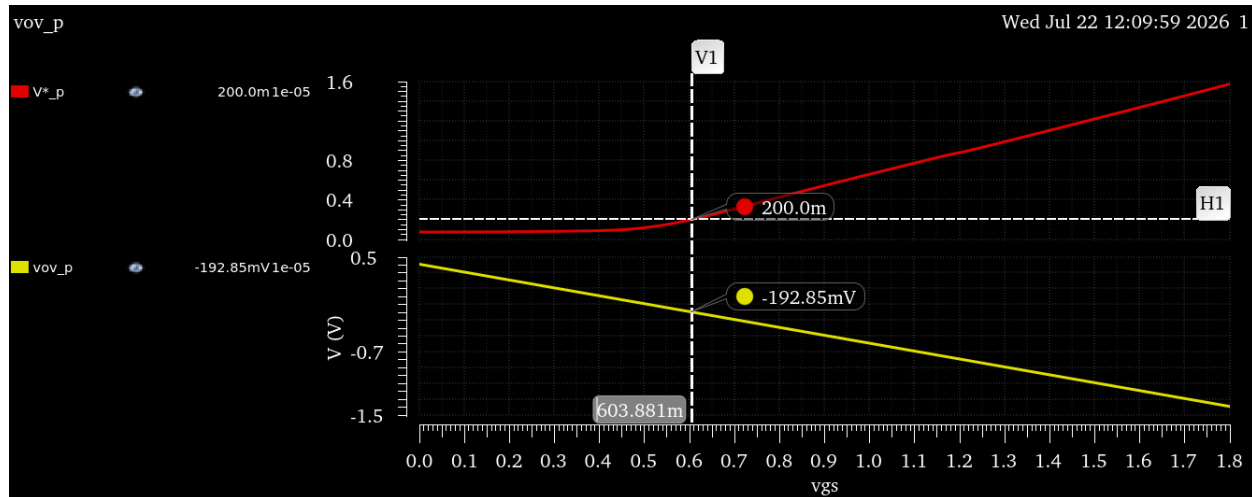


Figure 5: V^* & V_{ov} for pmos

- At $V^* = 200\text{ mV} \rightarrow V_{gs} = 603.881\text{ mV}$

$\rightarrow I_{dx} @ V_{gs} = 603.881\text{ mV}$

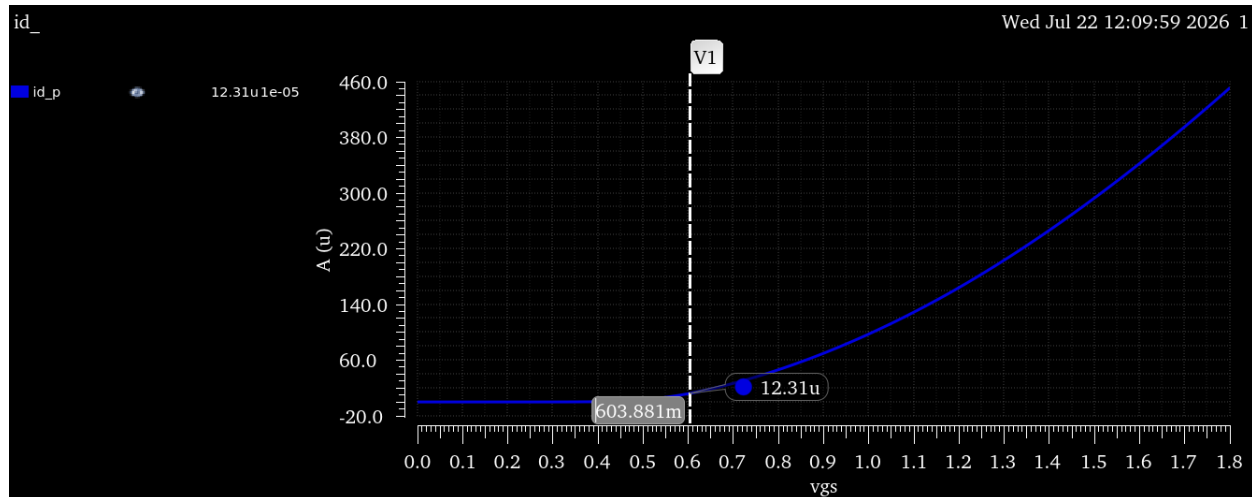


Figure 6: i_{dx} vs v_{gs} for pmos

→ Plot g_m , and g_{ds} vs V_G

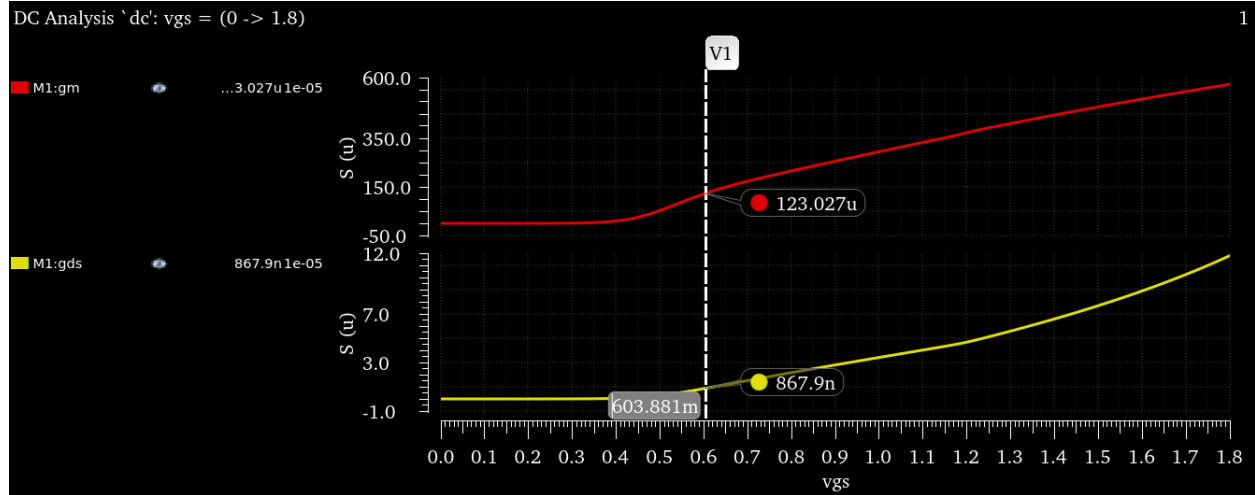


Figure 7: g_m & g_{ds} for pmos

- At $V_{gs}=603.881$ mV, from graph $g_m = 123.027$ μS & $r_o = \frac{1}{g_{ds}} = 1152.2$ $k\Omega$
- $I_{dx} = 12.31$ $\mu A \rightarrow W=10$ μm
- $I_d = 20$ $\mu A \rightarrow W_n = ?$
- $W_n = 10\mu m \times \frac{20\mu A}{I_{DX}} = 16.2469$ μm

Design Specifications Summary

Parameter	NMOS (@ $W = 10\mu m$)	NMOS final Design Value (@ Target $I_{DQ} = 20\mu A$)	PMOS (@ $W = 10\mu m$)	PMOS final Design Value (@ Target $I_{DQ} = 20\mu A$)
V^*	200 mV	200 mV	200 mV	200 mV
V_{GSQ}	605.378 mV	605.378 mV	603.881 mV	603.881 mV
V_{ovQ}	206.13 mV	206.13 mV	192.85 mV	192.85 mV
W	10 μm	4.2737 μm	10 μm	16.2469 μm
I_D	46.797 μA	20 μA	12.31 μA	20 μA
g_m	467.671 μS	467.671 μS	123.027 μS	123.027 μS



PART 2: Current Mirror

1) Schematic of the two CMs with DC node voltages clearly annotated at $V_{OUT} = V_{DD}/2$

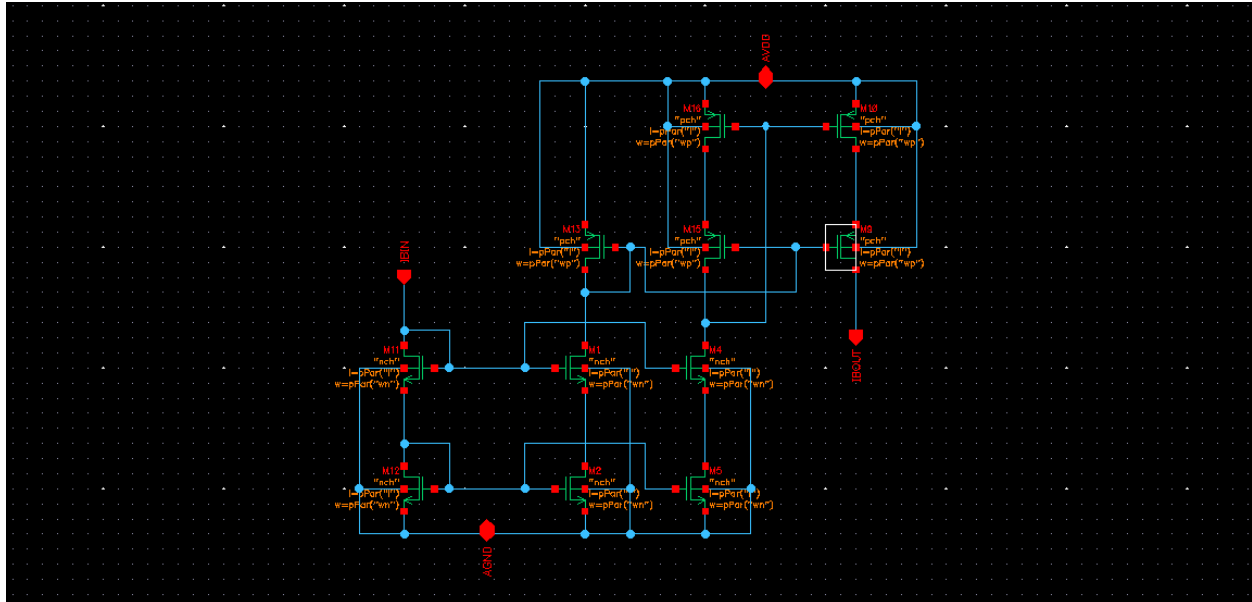


Figure 8: wide-swing current mirror

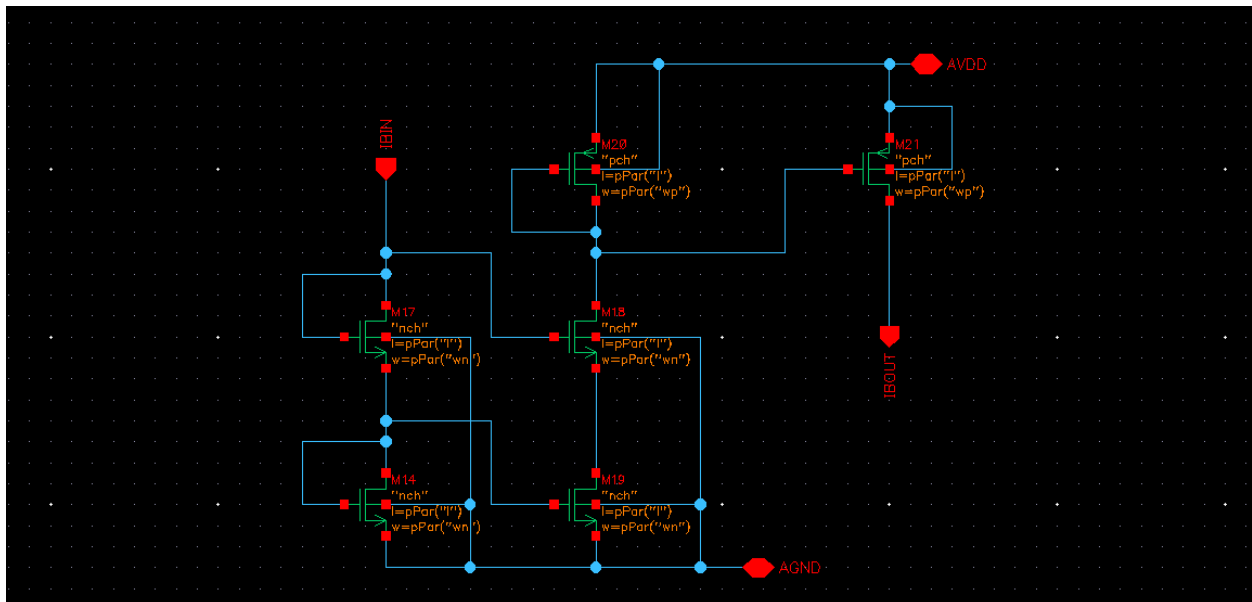


Figure 9: simple mirror

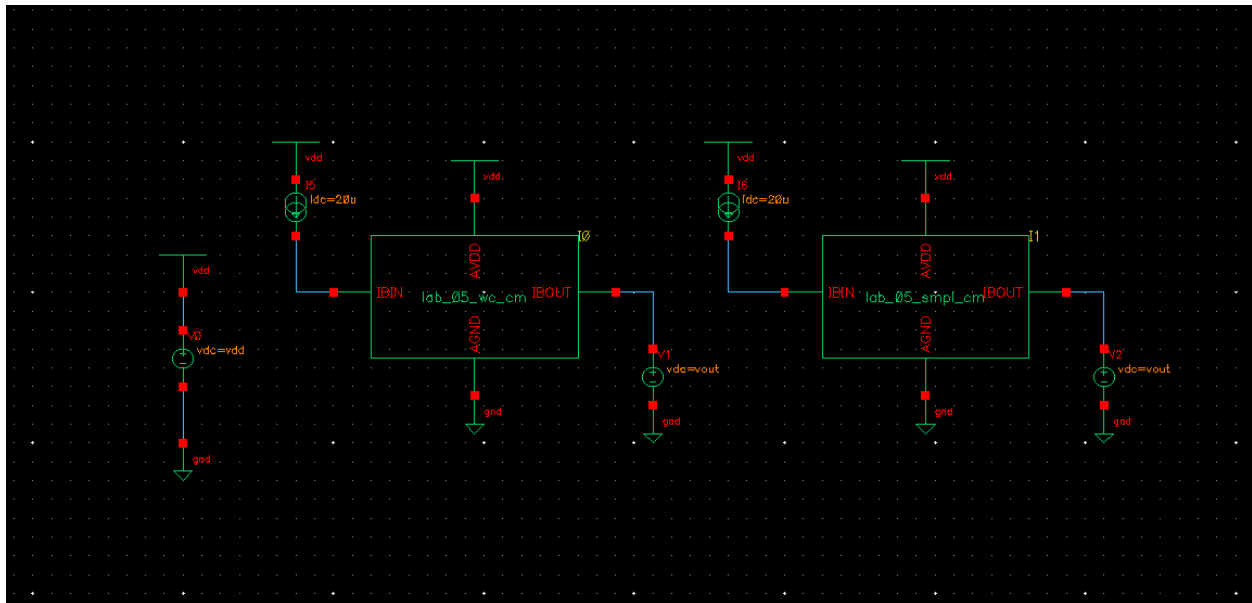


Figure 10:tb_schematic

2) Snapshots showing the following parameters for all transistors at $V_{OUT} = V_{DD}/2$.

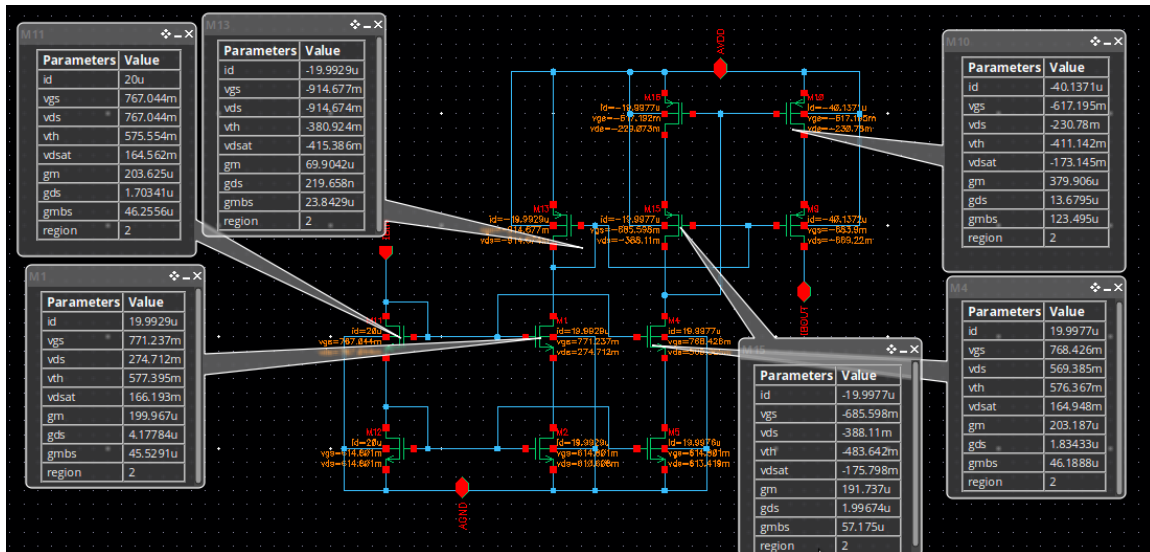


Figure 11: dc_op_wide_swing

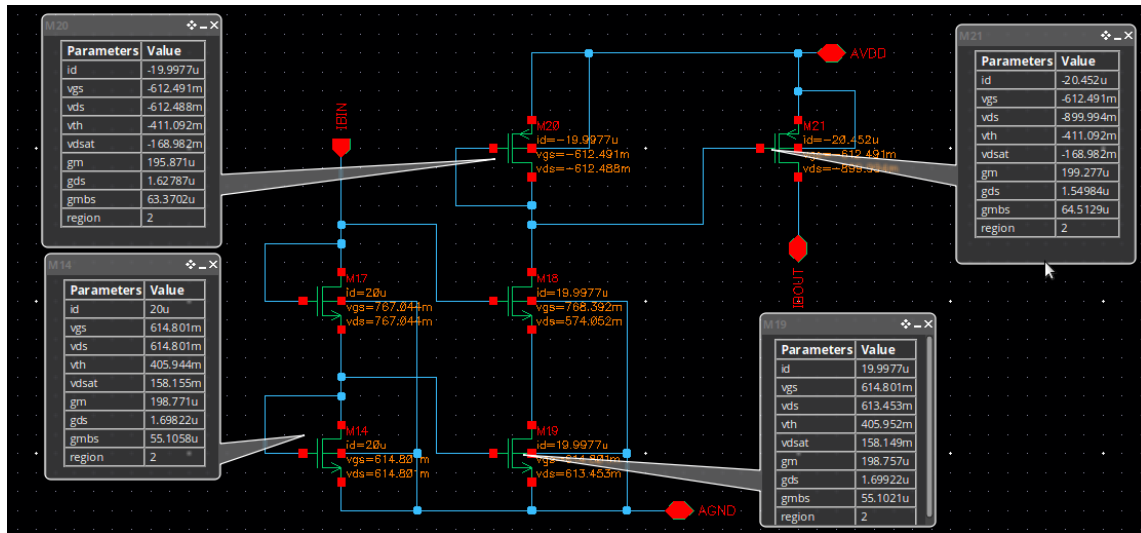


Figure 12: dc_op_simple_mirror

3) Check that all transistors have V_{GS} , g_m and g_{ds} as designed in Part 1.

es, the simulated values for V_{GS} , g_m , and g_{ds} closely match the calculated parameters from the Part 1 sizing chart, confirming the validity of the design methodology.

4) Are all transistors operating in saturation?

Yes, all transistors are operating in the saturation region (Region 2 in Cadence), as confirmed by the parameters table where $V_{DS} > V_{DSAT}$ for all devices.

5) IBOUT vs VOUT (DC Sweep)

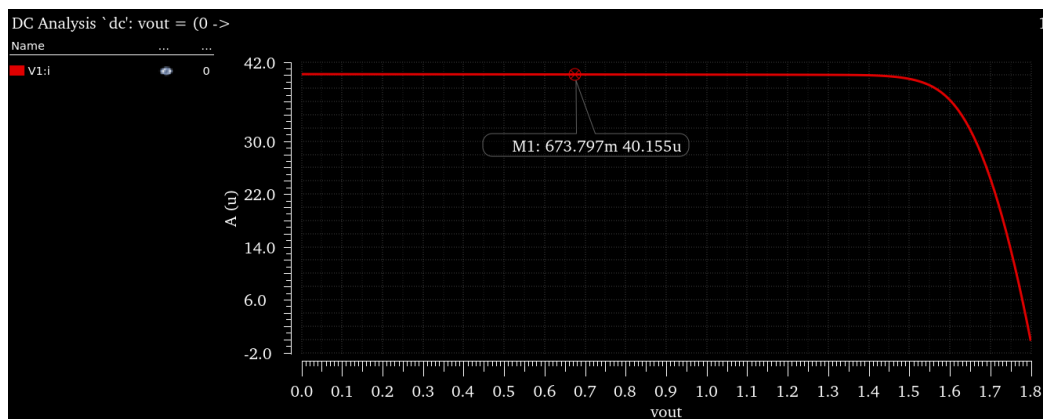


Figure 13: IBOUT vs VOUT for wide swing cm

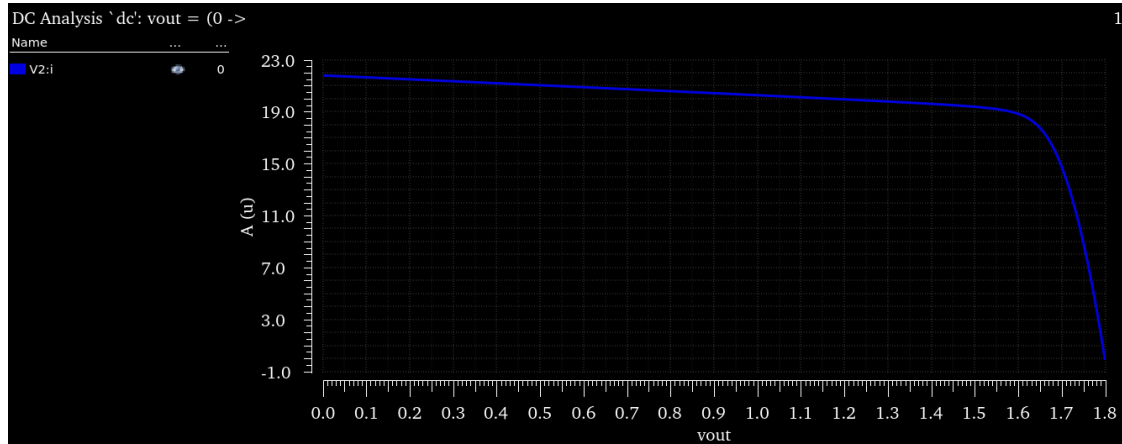


Figure 14: I_{OUT} vs V_{OUT} for simple cm

- Comment on the difference between the two circuits.

The Wide-Swing current mirror maintains a much flatter and more stable output current across the voltage sweep compared to the Simple current mirror, indicating a much higher output resistance.

- I_{OUT} of the simple CM is exactly equal to $I_{BIN} \times 2$ at a specific value of V_{OUT} . Why?

In the simple current mirror, the output current exactly matches the ideal ratio ($2 \times I_{BIN}$) when the output voltage V_{OUT} (which is V_{DS} of the output transistor) is exactly equal to the V_{DS} of the input diode-connected transistor. At this specific point, the channel-length modulation effect is identical for both transistors, eliminating the current mismatch.

6) Percent of Error in I_{OUT} vs V_{OUT}

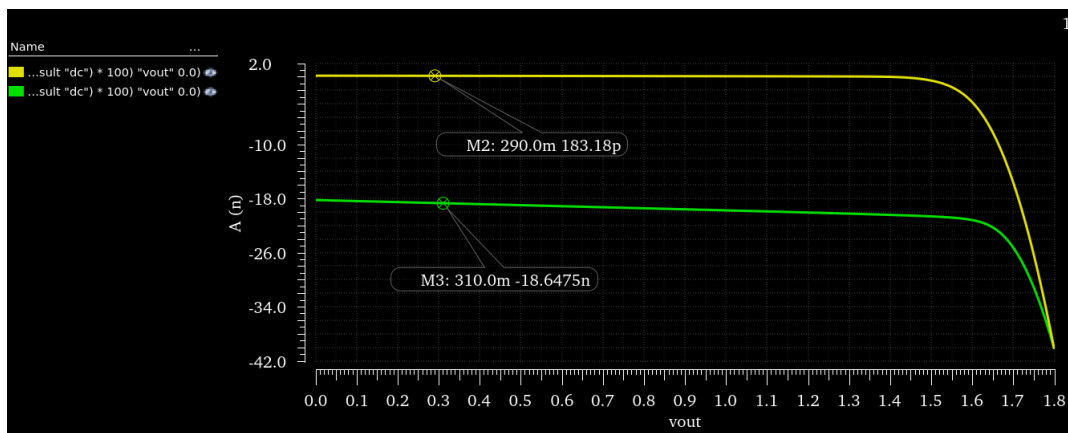


Figure 15: errors calculation



- Comment on the difference between the two circuits.

The percentage error in the Wide-Swing current mirror (yellow curve) is significantly lower and remains relatively constant over a wider compliance range. This is in contrast to the Simple current mirror (green curve), which shows a noticeable and steady change in error as V_{OUT} changes due to its much lower output resistance.

7) Rout vs VOUT (Log Scale)

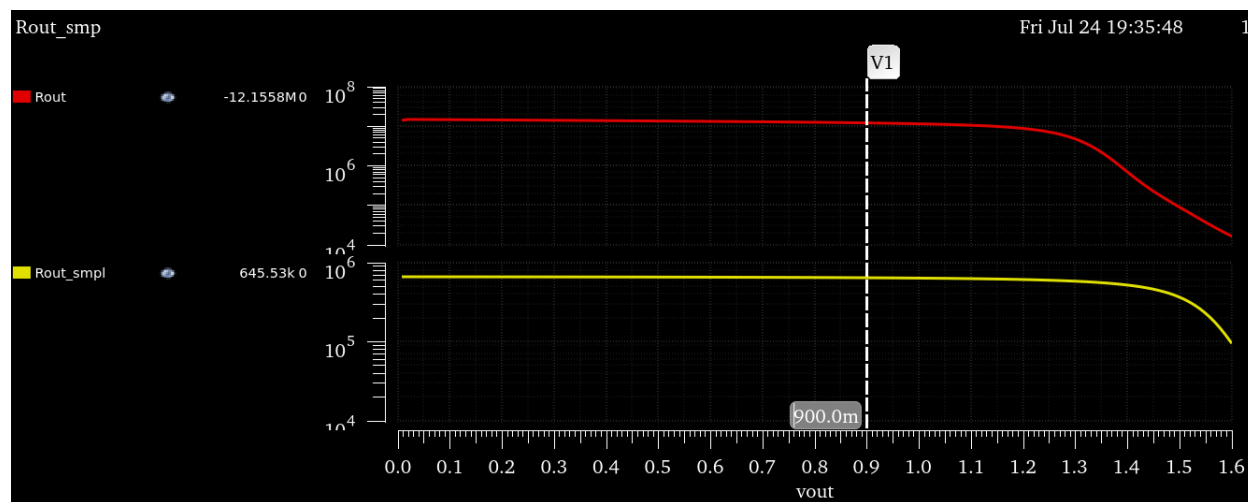


Figure 16: R_{out_wcm} vs. R_{out_smplcm}

- Comment on the difference between the two circuits.

The Wide-Swing cascode provides an output resistance in the Mega-Ohm range, which is orders of magnitude higher than the Simple current mirror (Kilo-Ohm range).

- Does R_{out} change with V_{OUT} ? Why?

Yes, R_{out} changes with V_{OUT} . This is due to the channel-length modulation effect. As V_{OUT} (which represents V_{DS}) increases, the depletion region at the drain expands, reducing the effective channel length (L_{eff}). This variation affects the output conductance (g_{ds}) and consequently changes R_{out} .



8) Analytical vs Simulation Table ($V_{OUT} = V_{DD}/2$)

- Simple Current Mirror Rout

$$R_{out} \approx \frac{1}{g_{ds(M21)}} = \frac{1}{1.54984 \times 10^{-6}} \approx 645.2 k\Omega$$

- Wide-Swing Rout

$$R_{out} \approx \frac{g_{m(M10)}}{g_{ds(M10)} \times g_{ds(M15)}} = \frac{379.9 \mu}{13.679 \mu \times 1.834 \mu} \approx 15.1 M\Omega$$

- Table 2: Comparison of Analytical and Simulated R_{out} at $V_{OUT} = V_{DD}/2$

Current Mirror Topology	Analytical Rout (Calculated)	Simulated R_{out} (From Plot)
Simple Current Mirror	645.2 k Ω	645.53 k Ω
Wide-Swing Current Mirror	15.1 M Ω	12.155 M Ω